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FEDERAL SPECIFICATION

SPREADERS, TIRE, PNEUMATICALLY OPERATED

This specification is approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers pneumatically operated tire spreaders for full circle and sectional spreading of tires to permit inspection, repair of the inside of the tire casings, and curing tube insertion operations.

1.2 Classification.

1.2.1 Types, styles, and sizes. The spreaders covered by this specification shall be of the following types, styles, and sizes, as specified (see 6.2):

Type I - Sectional.

Style A - Platform style, portable.
Size 1 - 7.50 - 10 through 16.00 - 25 tires.
Size 2 - 13.50 - 24 through 21.00 - 25 tires.

Style B - Pedestal style, stationary.

Type II - Full circle.

Style A - Pedestal style, with external spreader arms.
Size 1 - 7.50 - 10 through 16.00 - 25 tires.
Size 2 - All passenger, truck, and bus tires through 14.00 - 24.

Style B - Pedestal style, with internal spreader arms.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on date of invitation for bids or solicitation for offers, form a part of this specification to the extent specified herein.

Federal Specification

PPP-B-601 - Boxes, Wood, Cleated-Plywood

FSC 4910

AMSC N/A

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PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-T-60 - Tape, Packaging, Waterproof

Federal Standards

FED-STD-H28 - Screw-Thread Standards for Federal Services
FED-STD-123 - Marking for Shipment (Civil Agencies)
FED-STD-376 - Preferred Metric Units for General Use by the Federal Government

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes, are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications

MIL-P-116 - Preservation, Methods of
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-C-5501 - Caps and Plugs, Protective, Dust and Moisture Seal, General Specification for

Military Standards

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-889 - Dissimilar Metals
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; With Appropriate Test Methods

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Federal Regulation:

Occupational Health and Safety Administration (OSHA) 29 CFR 1910. 241-247 General Industrial Safety and Health Standards

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(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Fire Protection Association (NFPA)

NFPA 70 - National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 4.3 and 6.2), the contractor shall furnish one or more tire spreaders for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design. The spreaders shall be new and of the latest design meeting the requirements of this specification. The spreaders shall be pneumatically operated, floor mounted units for spreading tires in a vertical position. The spreaders shall not cause damage such as cutting, puncturing, or chafing of the tire tread, sidewall, bead, or the interior of the tire casing (see 4.6.1). The design shall be such that parts subject to wear, breakage, or distortion shall be accessible for adjustment, repair, or replacement without modification, distortion, or damage to the spreaders or any spreader part or component.

3.2.1 Type I. The spreader shall be a sectional-type spreader with external tire spreader arms to permit performance of sectional, or spot-repair, operations on the inside of the tire casing. The spreader shall permit rotation of the tire in the vertical position, without the operator lifting the tire, to perform the spreading operation on any desired section of the tire.

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3.2.1.1 Type I, style A. The spreader shall be a platform style, portable spreader capable of being used without being secured to a fixed surface. The spreader shall spread (open), without inverting, the tire casing and hold in that open position, within the size range specified in 3.5.1.1 and 3.5.1.2, to permit inspection of the tire and insertion or removal of curing bags.

3.2.1.2 Type I, style B. The spreader shall be a stationary unit and shall spread (open), inverting the tire casing and holding in that inverted position, within the size range specified in 3.5.1.3, to permit performance of sectional or spot-repair operations on the inside of the tire casing.

3.2.2 Type II. The spreader shall be a full circle spreader. Unless otherwise specified in 3.6.5, the spreader shall incorporate facilities for lifting the tire for positioning on the spreading arms. The spreader shall include provisions to permit tires in the spread (open) position to be rotated in the vertical position.

3.2.2.1 Type II, style A. The spreader shall be a stationary pedestal style with external style spreader arms, within the size range specified in 3.5.1.4 and 3.5.1.5, for the insertion and removal of curing tubes and for the inspection of the inside of tire casings.

3.2.2.2 Type II, style B. The spreader shall be a stationary pedestal style with internal style spreader arms, within the size range specified in 3.5.1.6, for spreading tires for inspection of the inside of tire casings.

3.2.3 Measurement system. The US Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the spreaders. In this specification, all measurements, dimensions, sizes, and capacities are given in the US system. These measurements may be converted to the SI system by using the conversion factors and methods specified in FED-STD-376.

3.2.4 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.2.5 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

3.2.6 Safety devices. Safety devices shall be provided for parts that present safety hazards to personnel. Safety devices shall not interfere with the operation of the spreaders or with the operator's performance.

3.2.6.1 Occupational safety standards. The product shall comply with OSHA standards set forth in 29 CFR 1910.241-247 that are applicable to spreaders.

3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or

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maintainability of the individual components or the overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.4 Construction. The spreaders shall be furnished complete so that, when installed and connected to the proper power source, they shall meet or exceed the requirements specified herein. Parts used in fabrication of the spreaders shall be new and free of defects. Part surfaces shall be clean and free of extraneous materials. External surfaces shall be smooth and free of sharp edges.

3.4.1 Castings and forgings. Castings shall be free from blowholes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from scale, inclusions, cold shuts, mismatching, sharp edges, or other defects. Castings and forgings shall be free from any properties adversely affecting their strength, durability, or suitability. Reclaiming or repair of major castings and forgings shall be permitted if the repair is not on a critical surface.

3.4.2 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain all requirements of the welded, brazed, or soldered parts. These operations shall not be employed as repair measures for defective parts.

3.4.3 Fastening devices. Screws, pins, bolts, and similar parts shall be installed with means for adjustment and for preventing loss of tightness. The methods for adjustment and for preventing loss of tightness shall be in accordance with accepted engineering standards and practices. All such parts when subject to renewal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

3.4.4 Lubrication. All bearings (except sealed, permanently lubricated type), mating gears, and other moving parts shall be provided means to assure adequate lubrication. Recirculating systems shall include filters which are cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid levels. Oil holes, grease fittings, and filler caps shall be located to be accessible. The manufacturer shall include and install all lubricants, fluids, greases, and other compounds in accordance with the best commercial practice of the producing industry (see 6.5). The lubrication systems shall be temporarily capped or stopped to prevent loss of lubricants during shipment. The manufacturer shall include a warning notice in the instruction manual and on the spreaders to remove the temporary caps or stoppers before start-up or use of the spreaders.

3.4.5 Electrical connections. Connections to screw-type terminals shall be mechanically secured with means to prevent loss of tightness. All other electrical connections shall be mechanically secured and soldered or brazed. Only rosin base fluxes shall be used in soldering operations.

3.5 Performance and product characteristics.

3.5.1 Capacity. The spreaders shall be capable of spreading tires within the following size capacities.

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3.5.1.1 Type I, style A, size 1. The spreader shall be capable of spreading tires within the minimum size range of 7.50 - 10 (4 ply) through 16.00 - 25 (20 ply), with a maximum spread width of not less than 16 inches.

3.5.1.2 Type I, style A, size 2. The spreader shall be capable of spreading tires within the minimum size range of 13.50 - 24 (20 ply) through 21.00 - 25 (24 ply), with a maximum spread width of not less than 36 inches.

3.5.1.3 Type I, style B. The spreader shall be capable of spreading tires within the minimum size range of 7.50 - 10 (4 ply) through 11.00 - 20 (12 ply), with a maximum spread width of not less than 16 inches.

3.5.1.4 Type II, style A, size 1. The spreader shall be capable of spreading tires within the minimum size range of 7.50 - 10 (4 ply) through 16.00 - 25 (14 ply), with a maximum spread width of not less than 16 inches.

3.5.1.5 Type II, style A, size 2. The spreader shall be capable of spreading all sizes of passenger, truck, and bus tires through 14.00 - 24 (20 ply) with a maximum spread width of not less than 16 inches.

3.5.1.6 Type II, style B. The spreader shall be capable of spreading tires within the minimum size range of 7.50 - 10 (4 ply) through 16.00 - 25 (14 ply), with a maximum spread width of not less than 16 inches.

3.5.2 Air pressure requirements. The spreaders shall comply with the following air pressure requirements.

3.5.2.1 Operating air pressure. The spreaders shall operate satisfactorily at an air pressure not to exceed 100 pounds per square inch gage (psig).

3.5.2.2 Excessive air pressure. The spreaders, and all components, shall be capable of withstanding, without damage or leakage, an air pressure of 250 psig.

3.5.3 Continuous duty. The spreaders shall be capable of withstanding, without damage, 300 continuous spreading cycle operations (full spread operating position and return). The type II, style A and B, spreaders shall also be capable of withstanding 300 continuous lifting cycle operations (full raised to lowered position).

3.5.4 Height and weight. The type I, style B, spreader shall not exceed 5 1/2 feet in overall height. The type I, style A, size 1, and type I, style B, spreaders shall not weigh more than 225 pounds.

3.5.5 Operational requirement. The spreaders shall be capable of being operated by one person, for all the operations specified herein, for the tire sizes specified in 3.5.1.1 through 3.5.1.6.

3.6 Detail of components. The spreaders shall consist of the principle components specified in 3.6.1 through 3.6.7.2.

3.6.1 Base. The spreaders shall incorporate a floor mounted base fabricated of metal and shall incorporate provisions for mounting to a fixed surface. The base shall maintain the spreader in a stable condition during the tire-spreading operation without being fastened to a fixed surface.

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3.6.2 Spreader arms. The spreaders shall incorporate spreader arms actuated by the pneumatic mechanism of 3.6.3 and embodying the detailed requirements specified for the respective type and style spreader in 3.6.2.1 and 3.6.2.2.

3.6.2.1 Type I, styles A and B. The spreaders shall have at least four external type tire spreader arms. The spreader arms shall engage the tire bead and shall function simultaneously and in conjunction with each other to perform the spreading operation. The arms shall be of a size to spread the tire beads of the tire sizes specified in 3.5.1.1 through 3.5.1.3. The spreader arms shall be arranged to not interfere with the performance of the functions specified herein. The distance between the adjacent spreader arm bead hook pads of the type I, style A, size 1 and type I, style B spreaders shall be not less than nine inches nor more than 14 inches. The distance between the adjacent spreader arm bead hook pads of the type I, style A, size 2, spreader shall be adjustable at any point within the minimum range of 10 through 19 inches. The spreader arms and bead hook pads shall be constructed to prevent disengaging from the tire bead or from injuring the tire sidewalls, bead, or casing interior when subjected to expanding pressure during operation of the spreaders. The spreader arm bead hook pads shall have rounded edges, which are free from sharp corners contacting the tire.

3.6.2.2 Type II, styles A and B. The type II, style A, spreader shall incorporate external spreader arms capable of holding and spreading the tire sizes specified in 3.5.1.4 and 3.5.1.5, and providing a full unrestricted opening of the tire in the vertical position. The type II, style B, spreader shall incorporate internal spreader arms capable of holding and full circle spreading, in the vertical position, the tire sizes specified in 3.5.1.6. The spreader arm hook pads shall engage the tire bead and shall function simultaneously to perform the tire spreading operation. The spreader arms shall be arranged to not interfere with the performance of the functions specified herein. The spreader arms and hook pads shall be constructed to prevent disengaging from the tire bead or from injuring the tire tread, sidewalls, bead, or tire casing interior during the tire spreading operation. The spreader arms shall be mounted on the pedestal to permit the tire in the spread condition to be rotated 360 degrees in the vertical plane. The spreader arm bead hook pads shall have rounded edges, free from sharp corners contacting the tire.

3.6.3 Pneumatic operating mechanism. The spreaders shall incorporate pneumatic operating mechanisms for transmitting the required force for accomplishing the tire spreading operations, as well as the tire lifting operation for type II, styles A and B, spreaders. The air actuated mechanism shall be capable of retaining an air pressure of 100 psig, for a period of 15 minutes, with a drop in air pressure of not more than 5 psig. The pneumatic actuating mechanism shall consist principally of an air cylinder, or cylinders, with operating controls, stops, and air connections embodying the detailed characteristics specified in 3.6.3.1 through 3.6.3.4.

3.6.3.1 Air cylinder. The air cylinder(s) shall be of the size and strength necessary for spreading and lifting the tire sizes specified when using the operating air pressure specified in 3.5.2.1. The internal working surfaces of the cylinder and piston shall be finished to the commercial tolerances required to retain the air pressures imposed during the spreading operations. The air piston, or pistons, shall have an automatic return when the air pressure is released.

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3.6.3.2 Air controls. The tire spreader mechanism shall incorporate an air control valve, or valves, for admitting, holding, and releasing the air from the air cylinders to control the action of the spreader arms. The type II, style A and B, spreader shall incorporate a valve for controlling the tire lift (see 3.6.5). The air control valve(s) shall be hand or foot operated and have at least three positions, "on", "release", and "neutral". The "on" and "release" positions of the control valves shall be distinctly marked. The operating controls shall be mounted in an accessible location on the spreaders to permit the simultaneous manipulation of the tire and operation of the controls. The air control valves shall be of the automatic shut-off type. The valve controlling the spreading arms shall be capable of holding the tire open at any point within the spread width range. The valve controlling the tire lift for type II, styles A and B, spreaders shall be capable of holding the tire at any point within the range of the lift.

3.6.3.3 Stops. The actuating mechanisms of the spreader arms and tire lift (for type II, styles A and B, spreaders, see 3.6.5) shall incorporate stops to prevent damaging the tire and to afford protection for the operator. The stops on the spreader arms shall prevent overtravel of the spreading width. The stops on the tire lift for type II, styles A and B, spreaders shall prevent overtravel of the lift piston. The stops shall be capable of withstanding the force imposed by the pressures specified in 3.5.2.1 and 3.5.2.2.

3.6.3.4 Air connections. The air connections between the air controls and air cylinder, or cylinders, shall be metal tubing or flexible hoses. If hoses are used, they shall be of flexible oil-resistant material. The hoses, with fittings, shall be capable of withstanding a hydrostatic pressure of 600 psig for a period of two minutes without damage such as bulging, fitting separation, leakage, splitting, bursting, or deformation. The air inlet fitting (air source to spreader) shall be a female thread, 1/4-18 National Pipe Taper. The air inlet fitting shall be in an accessible location and shall not cause the air supply line to interfere with the operation of the spreaders when connected to the fitting.

3.6.4 Tire support for type I, styles A and B. The spreaders shall incorporate a tire support for maintaining the tire in position on the spreaders, without the aid of the operator, after the tire has been positioned for spreading. The tire support shall embody the detailed requirements of 3.6.4.1 and 3.6.4.2.

3.6.4.1 Type I, style A. The tire support shall be composed essentially of at least two rollers. The support rollers shall permit the tire to be rotated when positioned on the spreaders to facilitate locating the tire prior to the spreading of a tire section. The rollers shall be permanently attached and shall operate freely without binding when used with the tire sizes specified herein.

3.6.4.2 Type I, style B. The tire support shall be composed essentially of a platen, or push plate, and at least two rollers. The support platen, or push plate, of the spreader shall press against the tire tread, causing the tire casing section being spread to be inverted and exposing the tire interior, facilitating inspection and repair of the tire. The support rollers shall permit the tire to be rotated, when positioned on the spreaders, to facilitate locating the tire prior to the spreading of a tire section. The rollers shall be permanently attached and shall operate freely without binding when used with the tire sizes specified herein.

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3.6.5 Tire lift for type II, styles A and B. Unless otherwise specified (see 6.2), the spreader shall be provided with a tire lift. The tire lift shall be actuated by the pneumatic mechanism (see 3.6.3). The tire lift shall incorporate facilities for lifting the tire sizes specified in 3.5.1.4 through 3.5.1.6 for positioning on the spreader arms (engaging the tire beads with the spreader arm hooks). The lift tire support shall be capable of being lowered to floor level to permit tires to be rolled onto the lift.

3.6.6 Tire loader for type I, style A. The spreader shall be provided with loading facilities to permit rolling tires into position on the spreader.

3.6.7 Inspection light for type I, style A, size 2; type I, style B; and type II, style A. The type I, style A, size 2 and type I, style B, spreaders shall be fitted with a shaded inspection light, located so that the light rays are directed on the complete tire section spread. The type II, style A, spreaders shall be fitted with two shaded inspection lights, with one light located on each side (inner and outer) of the tire and both directed on the lower (open) spread section of the tire. The inspection lights shall not interfere with the tire spreading operation. Unless otherwise specified (see 6.2), the inspection light fixtures shall accommodate not less than a standard 75 watt lamp and shall operate on 110 volt alternating current power source.

3.6.7.1 Switch. The inspection lights shall be provided with an accessible off-on control switch.

3.6.7.2 Power supply cord. Unless otherwise specified (see 6.2), the inspection light circuits shall be fitted with not less than ten feet of three conductor, flexible electrical cord, type SO heavy duty in accordance with NFPA 70. A three pronged molded plug cap, with a cord grip, shall be provided. The cord and cap shall be of the proper capacity for the current involved. The cord shall be reinforced or armored at the junction with the spreaders and shall be located to prevent the cord from interfering with the operation of the spreaders.

3.7 Painting. Unless otherwise specified (see 6.2), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting. At least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel), shall be applied.

3.8 Product identification.

3.8.1 Military agencies. The spreaders shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2), shall include the National Stock Number.

3.8.2 Civil agencies. The spreaders shall be marked in a legible and permanent manner with the manufacturer's name or trademark of such known character that the source of manufacturer can be readily identified.

3.9 Workmanship. Standards of workmanship shall assure that the spreaders shall have the stability, strength, durability, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to production items offered for acceptance under the contract. Quality conformance inspection shall consist of (a) through (d) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Performance test (see 4.6.1).
(c) Hose pressure test (see 4.6.5).
(d) Packaging inspection (see 4.7).

4.4.1 Sampling. Sampling for quality conformance inspection shall be in accordance with MIL-STD-105, using the following Inspection Levels and Acceptable Quality Levels (AQL):

Inspection or Test	Inspection Level	AQL
Product examination	II	1.0
Performance test	II	1.0
Hose pressure test	100 percent	-
Packaging inspection	S-4	4.0

4.5 Product examination. Visually and manually examine the spreaders to determine conformance with the requirements of 3.2 through 3.4.5 and 3.6 through 3.9. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from defects. Manual examination shall include the operation of movable parts by hand to assure proper functioning, measuring of dimensions, and weighing (as applicable). The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the spreaders have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection.

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4.5.1 Evidence of compliance with OSHA standards. Evidence of compliance with applicable safety and health requirements shall be reviewed (see 3.2.6.1 and 6.4).

4.6 Tests.

4.6.1 Performance test. The spreaders shall be installed and operated in accordance with the manufacturer's instructions. The largest and smallest tire size specified in 3.5.1.1 through 3.5.1.6, as applicable, shall be mounted on the spreader and each tire shall be spread at five different sections. During and upon completion of the spreading operation, the spreader and the tires shall be observed to determine compliance with the following:

(a) Design requirements (see 3.2).
(b) Tire capacity and bead spreader (see 3.5).
(c) Operating air pressure (see 3.5.2.1).
(d) Operational requirements (see 3.5.5).
(e) Spreader arms operation (see 3.6.2).
(f) Air controls accessibility and operation (see 3.6.3.2).
(g) Tire support operation (see 3.6.4).
(h) Tire lift operation (see 3.6.5).
(i) Tire loader function (see 3.6.6).
(j) Inspection light locations and lighting qualities (see 3.6.7).

4.6.2 Excessive air pressure test. The spreaders shall be subjected to a line pressure of 250 psig and while at this pressure the air control valves shall be operated, actuating the spreading mechanism for all types and the lifting mechanism for type II, styles A and B, 15 times through the full air piston(s) travel. During the test, the spreader shall be observed for damage and, upon conclusion of the test, the spreader shall be tested for leakage in accordance with 4.6.3 to determine conformance to 3.5.2.2.

4.6.3 Leakage test. Pressure gages shall be installed between the spreader arm valves for all types, and the tire lift control valves for type II, styles A and B and their respective air cylinders. With the spreader connected to an air source of not less than 100 psig, the air control valve, or valves, shall be opened separately permitting the air cylinder piston, or pistons, to be fully extended and entrapping an air pressure of 100 psig (as recorded on the test pressure gages). With the spreader arm valves and the lift-control valve (for type II, styles A and B only) closed for a period of 15 minutes, the pressure drop recorded on the pressure gages shall not exceed 5 psig (see 3.6.3).

4.6.4 Continuous duty test. The spreaders shall be installed and subjected to an operation of 300 spreading cycles for all types, and 300 tire lifting cycles for type II, styles A and B. The operating cycles shall be at 15 second intervals. Upon conclusion of the test, the spreaders shall be examined for damage and tested for performance and leakage in accordance with 4.6.1 and 4.6.3 to determine compliance with 3.5.3.

4.6.5 Hose pressure test. If air hoses are used in the pneumatic system, the air hoses, complete with fittings, shall be subjected to a hydrostatic pressure of 600 psig for a period of two minutes. During the test the hoses shall be observed, and upon conclusion of the test the hoses shall be examined to determine compliance with 3.6.3.4.

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4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Disassembly. The spreaders may be disassembled to the extent necessary to assure that all surfaces requiring a preservative are accessible for processing, to assure minimum cube for packing, and to permit reassembly without special skills or tools. Disassembly, when applicable, shall be as specified in MIL-STD-1186.

5.2 Preservation and packaging. Preservation and packaging shall be Level A or commercial, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Cleaning and drying. The spreaders shall be cleaned in accordance with process C-1 and dried in accordance with any applicable procedure of MIL-P-116.

5.2.1.2 Preservative application. Interior surfaces of the air cylinders shall be coated with type P-10, grade 10, preservative oil. The preservative shall be applied through the lubricating system, or by feeding into the air inlet while operating the spreader until preservative appears at the exhaust ports. Air inlets and outlets shall be closed with caps or plugs conforming to MIL-C-5501 or with tape conforming to PPP-T-60, class IV. All exterior, critical, or functional surfaces shall be preserved with type P-2 or P-11 preservative. The preserved surfaces shall be covered with barrier material conforming to MIL-B-121, grade A. Secure the barrier material with pressure-sensitive tape. All unplated or unpainted noncritical surfaces of the spreader shall be coated with type P-1 preservative. The preservatives and procedures for application shall be as specified in MIL-P-116.

5.2.1.3 Unit protection. The method of unit protection for the basic spreaders shall conform to method I of MIL-P-116. Small components which are coated with oil shall be covered with barrier material conforming to MIL-B-121, grade A. Parts which are susceptible to damage shall be packed in a fiberboard box conforming to PPP-B-636, class weather-resistant.

5.2.2 Commercial. Commercial preservation shall be as specified in ASTM D3951-82.

5.3 Packing. Packing shall be Level A or commercial, as specified (see 6.2).

5.3.1 Level A. Each spreader shall be packed in a box conforming to PPP-B-601, overseas type, or PPP-B-621, class 2. Blocking and bracing shall be in accordance with MIL-STD-1186. Closure and strapping shall be in accordance with the applicable container specification.

5.3.2 Commercial. Commercial packing shall be as specified in ASTM D3951-82.

5.4 Marking. Marking shall be military, commercial, or civil, as specified (see 6.2).

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5.4.1 Military. Military marking shall be as specified in MIL-STD-129.

5.4.2 Commercial. Commercial marking shall be as specified in ASTM D3951-82.

5.4.3 Marking for civil agencies. When specified (see 6.2), marking for shipment for civil agencies shall be as specified in FED-STD-123.

6. NOTES

6.1 Intended use. This specification covers sectional and full-circle spreaders used in field and base shops to facilitate tire repair operations.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, style, and size required (see 1.2.1).
(c) First article when required (see 3.1).
(d) Tire lift for type II spreader if different (see 3.6.5).
(e) Inspection light if different (see 3.6.7).
(f) Power supply cord if different (see 3.6.7.2).
(g) Painting and finishing if different (see 3.7).
(h) Military product identification if different (see 3.8.1).
(i) First article inspection if different (see 4.3).
(j) Preservation, packing, and marking if different (see 5.2, 5.3, and 5.4).
(k) Civil agency shipment marking when required (see 5.4.3).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.3.1 Repair part manuals. Repair part identification, in repair part manuals, should include the prime contractor's name, nomenclature, part number, equipment/part manufacturer (OEM - actual part manufacturer), Federal Supply Code for Manufacturers (FSCM), nomenclature, part number, and price of the part immediately after the prime contractor's part identification. High mortality rate items should be identified as shown in MIL-M-7298.

6.4 OSHA standards. Products which meet all requirements of this specification comply with the OSHA standards published prior to the date of this specification and identified in Section 3 (see 3.2.6.1).

6.5 Standard lubricants. The spreader manufacturer should specify and use to the maximum extent possible those lubricants (see 3.4.4) that are presently in the Department of Defense supply system. MIL-HDBK-113, MIL-HDBK-267, and MIL-STD-838 may be used as selection guides.

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MILITARY INTERESTS:

Custodians

Army - AL
Navy - YD
Air Force - 99

Review Activity

Air Force - 82

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

PREPARING ACTIVITY:

Army - AL

DoD Project 4910-0593

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